Climate Risk and Corporate Green Innovation: Evidence from Nearby Climate Disasters

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Abstract

We investigate the effects of heightened climate risk assessments on corporate green innovation. To isolate the effects of heightened climate risk assessments, we compare firms located in counties adjacent to, but not directly hit by, climate disasters with those located farther away. Using a stacked difference-in-differences design, we find that firms increase their green innovation following disaster events. Consistent with an information mechanism in which climate disasters reveal useful information that updates managers' beliefs about climate risk, the effects are stronger for firms in environments with higher levels of supply and demand for disaster-related information. Additional analyses indicate that the results are unlikely to be driven by economic spillovers from disaster zones or by salience-driven managerial overreaction. Overall, our findings provide new insights into when and how firms manage climate risk through green innovation.

JEL Classification: G32, G34, M41, M52, O32, Q54

Keywords: climate risk, climate disasters, green innovation, green patents, information dissemination, climate disclosures

"Climate change has become a defining factor in companies' long-term prospects. […] *The evidence on climate risk is compelling investors to reassess core assumptions about modern finance.* […] *In the near future – and sooner than most anticipated – there will be a significant reallocation of capital."*

Larry Fink's 2020 Letter to CEOs

1. Introduction

Climate risk has become an increasingly important determinant of firms' long-term cash flows.[1] The escalating frequency and intensity of extreme climate events have increased scrutiny of how firms assess and manage exposure to both the physical risks of climate change and the economic risks that arise from the transition to a lower-carbon economy. The financial stakes are substantial: the U.S. Fourth National Climate Assessment (2018) estimates that climate change could cost the U.S. economy hundreds of billions of dollars annually by the end of the century, a projection reinforced by the U.S. Office of Management and Budget (2021). Against this backdrop, green innovation emerges as a potentially important tool for firms to mitigate climate risk exposure, improve operational resilience, and position themselves for business opportunities associated with emerging green markets (Fink, 2022). However, despite the growing attention to this topic in recent years, we still lack a clear understanding of whether and how firms' assessments of climate risk translate into investment in green innovation.

There are compelling reasons to believe that firms facing heightened climate risk may increase their green innovation efforts to manage these risks. First, shifts in consumer demand toward products and services offered by companies that adopt climate-conscious policies can create revenue opportunities for investments in green technologies (Aragon-Correa and Sharma, 2003; Howard-Grenville et al., 2014; Li et al., 2024). At the same time, such shifts can also

[1] Climate risk refers to the potential negative effects of climate change on various aspects of the environment, businesses, society, and ecosystems (IPCC, 2020).

generate reputational damage, higher financing costs, and loss of market share, which negatively affect firms' bottom line if they fail to respond to climate risk (Matsumura et al., 2014; Engle et al., 2020; Krueger et al., 2020; Bolton and Kacperczyk, 2021; Baldauf et al., 2020; Li et al., 2024). Second, the physical risk from climate change could adversely impact firm operations. Extreme weather events can damage physical assets and increase operating costs (Dell et al., 2012; Kruttli et al., 2025b), creating financial incentives for firms to invest in green technologies that enhance operational resilience and reduce their long-term exposure to climate-related disruptions.

However, firms may not increase their green innovation activities for two reasons. First, investment theory suggests that uncertainty and irreversibility raise the option value of waiting, reducing firms' incentives to invest. Green innovation is often associated with a high degree of costs and uncertainty. Thus, firms may be reluctant to invest in green technology when it is unclear whether the expected returns justify the costs, particularly when such investments are partially irreversible (Arrow and Fisher, 1974; Bernanke, 1983; McDonald and Siegel, 1986; Bloom, 2009). Second, heightened risk may incentivize firms to adopt a conservative approach to avoid risk and prioritize stability. In such cases, they are more likely to focus on existing core business activities where they possess established expertise and competitive advantages, rather than diverting resources toward new initiatives like green innovation (Panousi and Papanikolaou, 2012; Julio and Yook, 2012; Gulen and Ion, 2016). Therefore, whether firms invest in green innovation in response to heightened climate risk is an empirical question.

Climate disasters are a natural, plausibly exogenous setting in which firms revise their assessments of climate risk. However, firms in directly hit areas also suffer from physical damage, financial losses, and recovery demands that can constrain innovative activity. To address this empirical challenge, we exploit precise disaster-county geographic information to study the green innovation of firms located in counties adjacent to climate disasters but *not* directly hit by them. These adjacent counties report no physical damage, financial losses, or human casualties from the disaster events, yet are exposed to elevated local information flows about the disasters, resulting in greater attention to these events than in more distant areas. Thus, this "near-miss" setting helps isolate the changes in climate-risk assessment prompted by climate disasters from contemporaneous shocks that directly affect firm resources and managerial priorities.

We obtain a list of major climate disasters in the U.S. and employ a stacked difference-in-differences design comparing the number of green patents from firms located in adjacent counties (treated firms) to those located in counties that are neither adjacent to nor directly hit by disasters (control firms). We define a county as adjacent to a climate disaster if the minimum distance between it and any directly hit county is within 100 miles (see Figure 1 for an illustration). We identify green patents using a combination of green patent codes developed by the Organisation for Economic Co-operation and Development (OECD), the World Intellectual Property Organization (WIPO), and the European Patent Office (EPO). We find that disaster-induced increases in climate risk assessments have a positive impact on corporate green innovation. In particular, treated firms produce approximately 2.33% more green patents following a climate disaster event. Economically, the magnitude of this effect is about 15% of the mean of green innovation, indicating a meaningful shift in green innovation behavior. Furthermore, we find no evidence of statistically significant pre-trends in green innovation, supporting the parallel trends assumption. The effects are incremental to controlling for time-varying firm characteristics and fixed effects, and are robust to alternative measures of green innovation and specifications. Together, these results are consistent with firms increasing their green innovation efforts to manage climate risk.

Next, we shed light on the economic mechanism behind the effects. We argue that climate disasters reveal useful information signals to firm managers, such as the realized severity and frequency of extreme climate events, the vulnerability of local infrastructure and business operations, as well as the pace and scope of recovery in these areas. Importantly, disasters also reveal how key stakeholders such as investors and consumers react to climate events through changes in attention, scrutiny, and economic behavior, thereby clarifying for firms how climate exposure translates into economic consequences. If climate disasters reveal relevant information that allows managers to update their beliefs about their firms' exposure to and vulnerability from climate risk, then the response in green innovation should be stronger in environments where disaster-related information is disseminated and processed more intensively. To examine this information mechanism, we conduct two sets of cross-sectional tests based on the local supply and demand for disaster-related information. On the supply side, we use media coverage of disaster-related news and find that the post-disaster increase in green innovation is more pronounced for firms located in states with high levels of disaster news coverage. Thus, this result supports the notion that the greater dissemination of disaster-related information from the media helps firms better assess their own climate risk exposure.

On the demand side, we proxy for stakeholder information demand using local Google Search volume for disaster-related terms and the number of disaster-related questions posed by analysts during conference calls. We find that the effect of climate disasters on green innovation is stronger for firms in areas with higher disaster-related search intensity and for firms subject to more analyst questioning. These results suggest that when stakeholders more actively seek disaster-related information, firms face stronger pressure to respond to climate risk, leading to a stronger green innovation response. Collectively, these tests provide support for the information mechanism underlying the effects of climate disasters.

To complement our analysis of innovation quantity, we also examine whether the quality of green innovation improves following climate disasters. We measure patent quality using two measures. The first is forward citations, which measure the invention's impact on future scientific and technological work. The second is the patent economic value from Kogan et al. (2017), which measures the benefits the invention is expected to generate for the firm. We find that the green patents produced by treated firms are higher in quality than those of control firms. Thus, firms are not merely increasing the number of green patents but also undertaking more impactful green innovation in response to climate risk, suggesting that post-disaster green innovation is unlikely to reflect greenwashing (Duchin et al., 2024).

Next, we conduct three additional analyses to further support our main findings. First, we verify that firms adjacent to directly hit counties experience heightened climate-risk assessments following climate disasters. To do so, we examine whether these firms increase their discussion of climate disasters in their annual reports. We find that firms become more likely to discuss climate disasters in their 10-K filings, and the frequency of disaster-related keywords in these filings also increases. These findings support the view that adjacent firms indeed increase their climate-risk assessments, even though they do not experience direct disaster impacts.

Second, climate disasters could generate spillover effects on adjacent counties and cause disruption in firm networks, which may affect the interpretation of our results. To address this concern, we first augment our baseline regression with a comprehensive set of county-level time-varying economic indicators (e.g., GDP, unemployment, job creation and destruction, etc.) to control for changes in adjacent counties' economic conditions. We find that the disaster effects on green innovation remain positive and significant, indicating that spillover effects on adjacent counties' economic environments do not explain the results. We next examine whether disruptions via peer networks could explain our findings by excluding treated firms with supply-chain or product-market linkages to firms located in directly hit counties. If the effects are driven by disruption in peer networks, then the effects should be muted after excluding treated firms with peers located in directly hit counties. However, we find that the main effects continue to be positive and significant after excluding these peer firms. Therefore, the evidence indicates that our main results are not explained by supply-chain or competitive shocks. Together, these findings are more consistent with the information mechanism in which firms respond to climate-risk information revealed by climate disasters.

In the final analysis, we assess the possibility that managers respond to disasters not because they update their beliefs about climate risk, but because they temporarily overreact to salient events (e.g., Dessaint and Matray, 2017). While our dynamic analysis does not indicate a reversal in green innovation response, we more directly distinguish between these interpretations by comparing green innovation responses for treated firms experiencing a climate disaster for the first time with treated firms that have previously experienced other disasters. If the effect were driven by salience-induced overreaction, we would expect strong responses only among first-time treated firms. However, the effects should be muted for non-first-time treated firms because repeated events lose their novelty and salience (Kahneman, 2011), and managers would have already learned from their previous overreaction. Empirically, we find that both groups exhibit similar increases in green innovation, and the difference between the two effects is statistically insignificant. Hence, this evidence is inconsistent with the notion that managers' green innovation response is driven by salience-induced overreaction.

Our paper makes several contributions to the literature. First, we add to the literature on the determinants of corporate innovation (Cohen and Klepper, 1996; Hall and Lerner, 2010; Brown et al., 2009; Aghion et al., 2013; Hirshleifer et al., 2013; Kim et al., 2025) and on how disclosure incentives shape innovation outcomes (e.g., Kim and Valentine, 2021, 2023; Dyer et al., 2024; Skinner and Valentine, 2025). We also contribute to the growing green innovation literature and respond to calls for more research on when and how firms are incentivized to invest in green innovation by showing that disaster-induced increases in climate risk assessments lead to stronger corporate green innovation responses (Glaeser and Lang, 2024; Acemoglu et al., 2012; Aghion et al., 2016; He and Qiu, 2025; Cheng et al., 2025).

Second, our paper also relates to the literature on the role of information dissemination and consumption in shaping corporate behavior. Prior work shows that external information supply such as media coverage can discipline managers and influence corporate policies (Dyck and Zingales, 2002; Core et al., 2008; Liu and McConnell, 2013). On the demand side, by actively seeking climate-related information, institutional investors and other stakeholders can exert pressure to alter firms' disclosure and real decisions (Krueger et al., 2020; Ilhan et al., 2023; Cohen et al., 2023; Drake et al., 2012; Yezegel, 2023). We contribute by providing evidence supporting an information mechanism in which the effects of disaster-induced climate risk assessments are more pronounced in environments with higher demand and supply of disaster-related information. These findings highlight an important policy implication: improving the dissemination of and access to climate risk information may help promote more meaningful green innovation.

Third, our paper also contributes to the literature on how climate disasters impact corporate and investor responses. Existing research shows that disasters shape financing conditions, such as higher loan spreads and implied volatility (Correa et al., 2025; Kruttli et al., 2025b), earnings effects in certain industries (Addoum et al., 2023), and stock price declines when key suppliers are hit (Barrot and Sauvagnat, 2016). Disasters also affect corporate policies such as cash hoarding (Dessaint and Matray, 2017), inventory stockpiling (Cho et al., 2025), and property insurance hedges (Aunon-Nerin and Ehling, 2008). We contribute by showing that when facing heightened climate risk, firms are incentivized to invest in green innovation as a way to manage climate risks.

Finally, a related strand of literature examines how disasters have heterogeneous effects on aggregate innovation in directly hit regions. These studies capture the combined influence of direct-hit consequences such as realized capital destruction, local disruptions, recovery dynamics, and policy responses that operate alongside changes in climate risk assessment (Le et al., 2024; Zhao et al., 2022; Miao and Popp, 2014; Ahmed et al., 2023; Kruttli et al., 2025a; Ma et al., 2023; Keding and Ritterrath, 2024; Moscona, 2025). In contrast, by focusing on adjacent firms, our near-miss design complements this literature by isolating the effect of disaster-induced climate-risk assessments on corporate green innovation in the absence of direct-hit consequences and highlighting an information channel through which this effect operates.

2. Hypothesis Development

Climate change exposes firms to risks that can materially affect their valuations, operations, and long-run viability. Prior research has examined how climate risk influences firm disclosures, operational decisions, and corporate policies (Li et al., 2024; Dessaint and Matray, 2017; Huang et al., 2022), as well as how it is reflected in expected returns and priced in capital markets (Bolton and Kacperczyk, 2021; Engle et al., 2020; Matsumura et al., 2014). However, one important yet relatively underexplored corporate response is green innovation—the development of technologies, processes, and products that reduce environmental impact, lower carbon footprints, or enhance resilience to climate-related shocks. Understanding whether and how firms invest in green innovation in response to climate risk is particularly relevant given its potential to address both the physical climate risks and the transition risks associated with the shift to a lower-carbon economy.

There are several reasons to expect a positive relation between climate risk and corporate green innovation. First, as stakeholders increasingly price and scrutinize firms' climate exposure, product-market demand can shift towards companies that proactively take climate-conscious actions. This shift creates revenue opportunities that increase the expected returns to green technologies (Aragon-Correa and Sharma, 2003; Howard-Grenville et al., 2014; Li et al., 2024). Even in the absence of such a change in demand, failing to respond to climate risk appropriately can result in reputational costs, higher financing costs, and loss of market share, adversely affecting firms' bottom line (Matsumura et al., 2014; Engle et al., 2020; Krueger et al., 2020; Bolton and Kacperczyk, 2021; Baldauf et al., 2020; Li et al., 2024). Second, climate change exposes firms to physical risks: extreme weather events can destroy physical capital, disrupt production, and increase operating costs (Dell et al., 2012; Kruttli et al., 2025b). These risks create direct financial incentives to invest in technologies that increase the resilience of infrastructure, improve energy and resource efficiency, and reduce exposure to future climate-related disruptions.

In our setting, nearby climate disasters play a specific and important role by providing useful information that can sharpen firms' climate-risk assessments. For example, extreme weather events bring climate risk to the forefront and reveal signals about the realized severity of climate shocks, the vulnerability of local infrastructure and business operations, and the pace and scope of recovery in similar local areas. Disasters can also reveal how various stakeholders react. For example, there could be increased attention and scrutiny on the extent to which firms take action to manage climate risk or contribute to a lower-carbon economy. Firms can also observe changes in consumer preferences due to heightened climate risk perception. Collectively, these signals help clarify for firms how climate exposure translates into economic consequences, informing managers' assessment of the expected costs and benefits of green innovation investments.

Additionally, while disasters are public events, their effects on climate-risk assessment are likely more pronounced for firms located close to the disaster zones. Proximity tends to intensify local information flows (e.g., local news supply and consumption) as well as attention, increasing the likelihood that managers revise their risk assessments and strategic responses. Therefore, we expect firms located near, but not directly hit by, disasters to update their climate-risk assessments more than firms located farther away, resulting in increased effort devoted to green technologies. Our main hypothesis, therefore, focuses on whether firms exposed to nearby climate disasters, which intensify climate-risk assessments, invest more in green technologies afterwards:

H1: Firms exposed to nearby climate disasters increase their green innovation.

However, theory does not unambiguously predict that heightened climate risk assessment will lead to more green innovation. Investment theory suggests that uncertainty and irreversibility of investment projects raise the option value of waiting (Arrow and Fisher, 1974; Bernanke, 1983; McDonald and Siegel, 1986; Bloom, 2009; Wang et al., 2026). Green innovation is typically costly, long-term, and subject to substantial technological and market uncertainty. When managers are uncertain about the distribution of future payoffs, they may prefer to postpone investments in green technologies. In addition, firms facing heightened risk may scale back investment and reduce R&D (Panousi and Papanikolaou, 2012; Julio and Yook, 2012; Gulen and Ion, 2016). From this perspective, climate disasters could prompt managers to adopt a more conservative approach to avoid risk and focus on existing core business activities rather than undertake new, risky green innovation projects. These competing forces suggest that the relationship between disaster-induced increases in climate risk assessment and green innovation is ultimately an empirical question.

3. Data and Descriptive Statistics

3.1. Data

3.1.1. Green Patent Data

We obtain patent data from Kogan et al. (2017), a comprehensive dataset that provides detailed information on patents granted by the United States Patent and Trademark Office since 1926. The dataset links patents to publicly listed firms and includes patent-level characteristics such as filing dates, grant dates, technological classifications (patent codes), and patent economic values derived from market reactions to patent grants.

To identify green patents, we first follow the recommendation of Favot et al. (2023) and combine three institutional taxonomies of environment-related technologies to determine whether a patent code is considered "green." These include the ENV-TECH list of green patent codes created by the Organisation for Economic Co-operation and Development (OECD), the International Patent Classification (IPC) Green Inventory list of green patent codes from the World Intellectual Property Organization (WIPO), and the Y02/Y04S tagging scheme developed by the European Patent Office (EPO). Although these schemes have been used individually in prior research (e.g., Cohen et al., 2020; Brown et al., 2022; Skinner and Valentine, 2025), Favot et al. (2023) show that combining all three sources to identify green patents substantially expands coverage and mitigates classification errors. Accordingly, we designate a patent code as "green" if it appears in any of the three lists. Next, we note that each patent often carries multiple patent codes in order to capture the full technological scope of an invention, which almost always spans more than one technical area. Therefore, we classify a patent as green if at least half of its codes are designated as green codes, ensuring that the patent's primary technological focus is environmentally oriented.

3.1.2. Climate Disaster Data

We obtain a list of major climate disasters in the United States from 1980 to 2019 from the National Centers for Environmental Information (NCEI).[2] According to NCEI (2025), these major climate disasters account for over 80% of the damage from all weather and climate events documented in the United States. Next, we match these disasters to county-level hazard data from the Spatial Hazard Events and Losses Database for the United States (SHELDUS) to identify the counties that suffer from any damages, casualties, or costs by a given disaster.[3]

To identify the counties adjacent to, but not directly hit by, the climate disasters identified above, we use the County Disaster Database provided by National Bureau of Economic Research (NBER) to classify a county into one of three categories: (i) directly hit counties, which include all the counties experiencing any damages, casualties, or costs, (ii) adjacent counties, which include those whose minimum distance between their geographical center and the center of any directly hit county is within 100 miles,[4] and (iii) other counties, which include all counties that are neither directly hit nor adjacent. Figure 1 provides an illustration of how counties are classified using Hurricane Michael in 2018 as an example.

Our final sample includes 129 major climate disasters, spanning six categories: tropical cyclones, winter storms, freezes, floods, severe storms, and wildfires. Appendix B reports the yearly distribution of disasters. The distribution indicates an increasing number of major climate

[2] A climate disaster is defined as a major disaster if overall damages/costs reached or exceeded $1 billion (including CPI adjustment to 2019). This restriction follows prior literature on climate disasters (e.g., Alok et al., 2020; Ludvigson et al, 2021; Kim et al., 2025)
[3] SHELDUS collects damages, casualties, and costs data from various sources and aggregates to disaster-county-level.
[4] The distance is calculated using the Haversine formula based on center points in the geographic area. In Section 5.4, we assess the robustness of the main results using alternative definitions of adjacent counties: those within 50 or 150 miles of directly hit counties, those that are bordering counties (i.e., share the same border with directly hit counties), and those that are among the top 10 nearest counties to the inverse hyperbolic sine (IHS)-transformed ly hit counties.

disasters over the years, consistent with the documented global rise in extreme climate events (IPCC, 2021).

3.1.3. Other Data

We obtain firm accounting data from the CRSP/Compustat Merged database and institutional ownership data from the Thomson Reuters 13F database. For supplementary analyses, we combine several additional data sources: textual content from firms' 10-K filings from EDGAR, conference call transcripts from Capital IQ, supply chain data from Compustat Segments, and product market peer data from the Hoberg-Phillips Data Library. We further obtain state-level data on demand and supply of disaster-related information from Google Trends (Google search index) and RavenPack (news coverage), respectively. Local economic data at the county level (such as GDP, unemployment, etc.) is obtained from various government agencies, namely the Bureau of Economic Analysis (BEA), the Bureau of Labor Statistics (BLS), and the U.S. Census Business Dynamics Statistics (BDS).

3.2. Sample Selection and Descriptive Statistics

Because different climate disasters occur in different years through the sample period, we use a stacked difference-in-differences (DiD) event-study design. For each disaster, we consider an event window spanning five years before and after the disaster, denoted as [-5, +5], with the disaster year designated as year 0. We consider firms located in the adjacent counties as treated firms, and the firms located in the other counties as control firms.[5] To illustrate, the map in Figure 1 distinguishes among directly hit counties (red), adjacent unimpacted counties (gray), and other control counties (unshaded). Firms in the gray counties are treated firms, whereas firms in the unshaded counties are control firms. Following the literature on the treatment effect of repeated

[5] Firm location information is obtained from historical 10-K filings.

non-absorbing shocks (Cengiz et al., 2019; Dube et al., 2025; Chaisemartin and D'Haultfœuille, 2024), we require that control firms have never been treated at any time during the [-5,+5] window.[6] Each cohort (i.e., stack) corresponds to a disaster, consisting of treated and control firms associated with said disaster.

We then append the stacks together to obtain the initial sample. Because our analyses focus on firms with the capacity to generate innovation, we restrict the sample to firms that have filed at least one patent in their corporate history. We apply the inverse hyperbolic sine (IHS) transformation to all count-like variables, such as our main dependent variable of interest (*GreenPatent*), to account for frequent occurrences of zeros and the skewed distribution of these variables.[7] We drop observations with missing values of variables used in the main analysis. All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. The resulting main (stacked) sample consists of 461,153 disaster-firm-year observations from 1979 to 2021.

Table 1 reports the descriptive statistics for the variables used in our main sample. Appendix A provides definitions of all variables. The mean of the *Treat* variable is 0.309, indicating that approximately 30.9% of the observations in our sample correspond to firms in the counties adjacent to the directly hit areas. The mean of the *Post* variable is 0.547, consistent with our defined event window. The mean of our dependent variable of interest, *GreenPatent*, is 0.155, suggesting that the average number of green patents filed per year is approximately 0.156. The

[6] Recent developments in econometrics (e.g., Dube et al. 2025; Chaisemartin and D'Haultfœuille 2024) show that when treatment shocks, such as climate disasters, are frequently repeated and non-absorbing (i.e., treatment can occur multiple times and does not permanently switch a unit into a treated state), imposing the conventional not-yet-treated requirement often used in stacked DiD would significantly shrink and distort the control pool, mis-specify treatment histories, and require unnecessarily strong identifying assumptions. Instead, firms treated outside of a given event window can still serve as valid controls, provided that the parallel trends assumption holds locally within that window.
[7] The IHS transformation is superior to the log-one-plus transformation in handling skewed data with frequent occurrences of zeros (Glaeser and Omartian, 2022; Kim, Shi, and Verdi, 2025; Kim and Valentine, 2025). This transformation yields coefficient estimates that can be approximately interpreted as percentage changes.

low average reflects the fact that most firms do not file any green patents in a given year. The standard deviation of *GreenPatent* is 0.561. The skewness of *GreenPatent* is consistent with prior evidence in the innovation literature (e.g., Cheng et al., 2025; Kim and Valentine, 2025).

Turning to firm characteristics, the mean and median of firm size (IHS transformation of total assets) are 6.339 and 6.150, respectively. The leverage ratio has a mean and median of 0.173 and 0.111, respectively. The average (median) firm has a return-on-assets of -0.081 (0.021) and a book-to-market ratio of 0.504 (0.395). The average IHS transformation of firm age is 3.265, suggesting that the average firm in our sample is approximately 13 years old. These characteristics, along with other controls, are generally comparable to those found in recent studies on corporate innovation (e.g., Huang et al., 2021; Cheng et al., 2025; Valentine et al., 2025).

4. Main Analyses

4.1. Main Results: The Effects of Increased Climate Risk Assessments on Green Innovation

We begin our analyses by examining hypothesis H1, which tests whether disaster-induced increases in climate risk assessments influence corporate green innovation for firms located in nearby counties. To test this hypothesis, we use a stacked DiD regression model:

$$
\begin{aligned}
GreenPatent_{i,t} = \alpha_1 + \beta_1\, Treat_i \times Post_{i,t} &+ \beta_2\, Size_{i,t} + \beta_3\, Leverage_{i,t} + \beta_4\, ROA_{i,t} \\
&+ \beta_5\, Age_{i,t} + \beta_6\, Cash_{i,t} + \beta_7\, BTM_{i,t} + \beta_8\, Financial\ Constraint_{i,t} \\
&+ \beta_9\, HHI_{i,t} + \beta_{10} CAPX_{i,t} + Cohort\text{-}Firm\ FE + Cohort\text{-}Year\ FE + \varepsilon_{i,t}, \qquad (1)
\end{aligned}
$$

where subscripts i, t denote firm i and year t, respectively. The dependent variable of interest is *GreenPatent*, the IHS-transformed number of green patents filed by the firm in a given year. *Treat* is an indicator equal to one for treated firms located in adjacent counties, and zero for control firms located in other counties not adjacent to nor directly hit by natural disasters (see Section 3.2 for additional details). The primary coefficient of interest is β_1 of the interaction term *Treat* × *Post*, which estimates the differential change in green innovation for treated firms relative to control firms following a disaster. A positive coefficient β_1 would support the hypothesis that increases in climate risk assessments lead to more green innovation.

We include a set of control variables to account for time-varying firm characteristics that may influence corporate innovation. For example, *Size* and *Age* capture scale and life-cycle effects that systematically relate to firms' R&D and innovation activity (Cohen and Klepper, 1996; Hirshleifer et al., 2013). *ROA* and *Cash* proxy for internal financing capacity, while *Leverage* and *Financial Constraint* capture financing frictions that can restrict innovative investment (Hall and Lerner, 2010; Brown et al., 2009). We also include book-to-market ratio (*BTM*) to capture variation in growth opportunities. We control for industry competition using the Herfindahl-Hirschman Index (*HHI*) and institutional investors' influence on firm investment policy (*Institutional Holding*) (Aghion et al., 2013). Finally, we include capital expenditure (*CAPX*) to control for overall capital investment patterns that may complement or crowd out innovation investments.

Following the recent literature on stacked DiD (Cengiz et al., 2019; Baker et al., 2022), we include cohort-firm fixed effects and cohort-year fixed effects, where each cohort corresponds to a climate disaster, to absorb time-invariant firm heterogeneity and aggregate time trends within each disaster. These fixed effects fully absorb the main effects of *Treat* and *Post*, rendering their explicit inclusion redundant. Standard errors are clustered at the firm level.

Table 2 presents the results of our baseline estimation of Equation (1). Column (1) reports the regression without controls, where the coefficient on the interaction term *Treat* × *Post* is positive and statistically significant at 0.0189 (*t*-stat = 2.53). In Column (2), we include the full set of firm-level controls. The coefficient of interest continues to be positive at 0.0233 and statistically significant at the one percent level (*t*-stat = 2.86). The results suggest that heightened climate risk assessment induced by climate disasters leads to an increase in green innovation. In terms of economic magnitude, the coefficient of 0.0233 suggests that, relative to control firms, treated firms increase their green innovation activity by approximately 2.33% in the post-disaster period. This magnitude is roughly equal to 15% of the unconditional mean of *GreenPatent*, indicating a meaningful increase in green innovation.

Next, we assess the parallel trends assumption underlying the DiD estimation. To do so, we estimate a dynamic version of Equation (1) by replacing the *Post* variable with separate event-time indicators for each year in the [-5, +5] window relative to the disaster, omitting *t*-1 as the benchmark. Figure 2 plots the point estimates and 95% confidence intervals. As shown in the figure, the coefficients in the pre-period are small in magnitude and statistically indistinguishable from zero. In contrast, the coefficients on the post-treatment lags are positive and statistically significant. This dynamic pattern is consistent with the notion that climate disasters serve as information shocks, allowing managers to more accurately assess their firms' exposure to climate risk and thereby stimulating long-term investment in green innovation.

Our design focuses on adjacent counties to isolate the innovation response to disaster-induced revisions in climate-risk assessment from direct operational and financial consequences of being hit. For completeness, also examine firms located in directly hit counties. Table C.1 in Appendix C reports the results. We replace the *Treat* variable in Equation (1) with *DirectHit*, an indicator variable equal to one for firms located in counties that are directly hit by a given climate disaster, and zero for those located in counties that are neither directly hit nor adjacent. We find that the coefficient on *DirectHit* × *Post* is statistically insignificant and economically close to zero (e.g., -0.0003 in Column (2)). This null result is consistent with the idea that directly impacted firms face competing economic forces: although disasters may heighten climate risk assessments, these firms also experience immediate financial losses, physical damage, operational disruptions, and recovery demands that hinder their ability to invest in green innovation.

4.2. Information Mechanism – Supply and Demand of Disaster-Related Information

Our hypothesis posits an information mechanism: nearby climate disasters are not only physical events, but also serve as informational shocks that update managers' beliefs about climate risks. Whether firms ultimately increase green innovation in response to climate disasters depends on the effectiveness of this information flow (i.e., the extent to which disaster-related information is disseminated and consumed). Therefore, we expect the treatment effects to vary cross-sectionally with the information environment surrounding the disasters. We test this prediction by examining both the local supply of and demand for disaster-related information.

4.2.1. Supply: News Coverage

We first investigate the supply of information. The accounting and finance literature documents a pivotal role of the media in transmitting and shaping information environments (Bushee et al., 2010; Fang and Peress, 2009; Fritsch et al., 2025). Media outlets often provide extensive coverage and analysis of the causes and consequences of large climate disasters. This flow of information helps firms and stakeholders better understand the scope, frequency, and intensity of climate risk in a given region, and in turn allows them to reassess the relevance of observed disasters to their own climate risk exposure. Additionally, extensive news coverage can heighten managerial attention to relevant information, ensuring that managers react to the information signal (Huberman and Regev, 2001). Thus, we expect that the corporate green innovation response in the post-period is stronger when there is greater disaster-related news coverage.

We measure the extent of the local supply of disaster-related news using RavenPack data. We identify disaster-related articles based on whether their headlines contain at least one disaster-related keyword provided in Appendix D. To avoid endogeneity concerns in which contemporaneous disaster coverage is related to the disaster in question, we use a firm's pre-treatment exposure to disaster-related coverage in the firm's headquarters state.[8] Specifically, we augment Equation (1) by adding the interaction between *Treat* × *Post* and *PreDisasterNews*, which is defined as the decile rank of the average annual count of disaster-related news articles in the firm's location state in the pre-period. The results are reported in Table 3.

Because the news coverage data from RavenPack starts in 2000 and we also require at least three years of pre-treatment data to calculate the average news coverage, the sample size in this analysis decreases to 195,897.[9] Therefore, we first replicate the main result and report the estimates in Column (1). The coefficient on *Treat* × *Post* is positive and statistically significant. More importantly, in Column (2), the coefficient of interest, *Treat* × *Post* × *PreDisasterNews*, is positive and statistically significant at the one percent level, suggesting that the main effect is more pronounced for firms located in states where there is a high level of disaster-related news coverage.

To ensure this result is not an artifact of the state size or economic activity, we employ two alternative measures of pre-period news coverage. The first is *PreDisasterNewsPerCapita*, which is the decile rank of a firm's average annual count of disaster-related news articles scaled by the population of the firm's state in the pre-period (Column (3)). The second measure is

[8] We measure news coverage at the state level because RavenPack does not provide data at the county level.
[9] We impose this restriction to mitigate measurement errors and ensure that our measurements capture the persistent, ex-ante information environment rather than transitory fluctuations. Our results are robust to removing this restriction.

PreDisasterNewsPerGDP, defined as the decile rank of a firm's average annual count of disaster-related news articles scaled by the GDP of the firm's state in the pre-period (Column (4)). Across both specifications, the coefficients on the triple interaction terms remain positive and statistically significant. Overall, these findings support the notion that a high supply of disaster-related information facilitates managers' ability to assess their firms' climate risk exposure, leading to a more pronounced green innovation response.

4.2.2. Demand: Google Search Index & Analyst Participation in Conference Calls

We next investigate the demand for information. Christensen et al. (2021) note that corporate environmental policies are increasingly shaped by the monitoring and pressure from diverse stakeholders. For example, disasters can heighten public concern about climate change and increase stakeholder demand for environmentally responsible behavior. Consumers may favor green products and services while boycotting companies perceived to be unfriendly to the environment. Investors may intensify their scrutiny of firms' climate risk management strategies. These external pressures generate stronger incentives for firms to act on climate risk. Hence, we expect that managers are more likely to allocate resources to green innovation when they face stronger pressure from stakeholders.

Because stakeholder pressure often materializes through heightened attention and information demand, we test this prediction using variation in information acquisition. In particular, we focus on two groups of stakeholders: the general public (which represents social pressure) and financial analysts (which represent capital market pressure). Following Da et al. (2011) and Drake et al. (2012), we proxy for public demand for disaster-related information using Google Search volume. Similar to before, we use pre-period measures to mitigate endogeneity concerns. We define our cross-sectional variable, *PreGoogleSearch*, as the decile rank of a firm's pre-treatment average level of disaster-related Google searches, calculated as the average Google search index of disaster-related keywords in the firm's location state in the pre-period.[10] The Google search index data begins in 2004. We also require at least three years of pre-treatment data to calculate the average public information demand. The sample size reduces to 195,520 observations. We report the results in Table 4. Column (1) replicates the main result as the benchmark. In Column (2), the coefficient on the triple interaction term, *Treat* × *Post* × *PreGoogleSearch*, is positive and statistically significant. The results, thus, suggest that public demand for disaster-related information compels stronger green innovation responses.

Second, we examine the capital market's demand for disaster-related information using analyst questions during conference calls. Yezegel (2023) and Ashraf et al. (2024) document that analysts play a critical role in eliciting value-relevant information from management. Consistent with this view, we argue that analysts actively questioning managers about disaster-related issues represents higher information demand from capital markets. We obtain conference call transcripts from Capital IQ starting in 2006, and require three years of pre-treatment analyst data to calculate the number of analysts asking disaster-related questions and the number of disaster-related keywords mentioned by analysts during the Q&A sessions. We define *PreAnalystQuestion* as the decile rank of the firm's average number of analysts who ask disaster-related questions in the pre-period, and *PreAnalystWord* as the decile rank of the firm's average total number of disaster-related keywords from analysts' questions during conference calls in the pre-period. The sample requiring non-missing values of these variables has 125,796 observations. In Column (3) of Table 4, we replicate the main result as a benchmark. Importantly, as shown in Columns (4) and (5), the coefficients on the triple interaction terms are positive and statistically significant, indicating that

[10] Because county-level Google search index is not available, we use state-level data.

explicit information demand from sophisticated market intermediaries incentivizes managers to act on climate risk, resulting in a stronger green innovation response.

Collectively, the results from Tables 3 and 4 suggest that firms increase their green innovation efforts when the demand and supply of disaster-related information are high, supporting our postulated information mechanism. That is, nearby climate disasters reveal useful information to managers, and the effects of disaster-induced climate risk assessments are more pronounced in environments where such information is disseminated and consumed more intensively.

4.3. Green Innovation Quality

While our baseline results in Section 4.1 document an increase in the quantity of green patents, the innovation literature notes that the economic significance of innovation depends not only on the number of patents but also on their technological impact and economic value (e.g., Hall et al., 2005; Kogan et al., 2017). Moreover, a potential concern is that the observed increase may reflect greenwashing, a phenomenon in which firms strategically produce filings designed to virtue signal environmental commitment rather than demonstrate substantive technological advancement (Duchin et al., 2024). If the observed effect is primarily symbolic, we would not expect climate disasters to lead to higher technological impact or economic value among these patents. To address these issues, we extend our analysis to examine the quality of green patents produced after climate disasters. We gauge the quality of a patent along two dimensions: its scientific value and its economic value. A patent's scientific value refers to the extent to which the patented invention contributes to subsequent scientific and technological developments. Its economic value, however, refers to the financial returns or commercial benefits the patent is expected to generate for the firm.

To capture the scientific value of a patent, we use forward citations. Intuitively, if a patent pushes the frontier of technology advancement, then we expect it to be cited more by subsequent inventions. We define *GreenCitation* as the average number of forward citations among the green patents filed by the firm in a given year, with the IHS transformation applied. We re-estimate Equation (1) with *GreenCitation* as the dependent variable. The results are reported in Table 5. Across both specifications, the coefficient on the interaction term *Treat* × *Post* is positive and statistically significant. Economically, the estimate from Column (2) suggests that, relative to control firms, treated firms see an increase of roughly 4.4% in forward citation counts for green patent filed following a climate disaster, which is roughly 23.8% of the mean value of *GreenCitation*.

Next, we measure the economic value associated with these green innovations. We utilize the measure developed by Kogan et al. (2017), which estimates the economic value of a patent based on the stock market reaction around the patent's grant date. We define *EconomicValueGreenPatent* as the average economic value of the green patents filed by a firm in a given year. Re-estimating Equation (1) with *EconomicValueGreenPatent* as the dependent variable, we report the results in Columns (3) and (4) of Table 5. The coefficient of interest is positive and statistically significant at the one percent level. The estimate in Column (4) suggests an increase of 0.1192. Economically, this increase is roughly 15.3% of the mean value of patent economic value.

Overall, the results in Table 5 indicate a significant increase in the quality of green patents filed by treated firms in the post-disaster period, both in terms of their scientific impact and economic value, relative to those filed by control firms. These findings suggest that the increase in green innovation is not merely a rise in the number of filings but reflects patents that are technologically impactful and viewed by the capital market as economically valuable.

5. Additional Analyses

In this Section, we conduct a series of additional tests to further support our main findings and to assess alternative explanations. In particular, we verify that adjacent firms exhibit heightened climate-risk assessments following nearby disasters. We then examine whether our results could be driven by potential spillover effects on adjacent counties or by economic linkages between treated firms and those in disaster zones. Furthermore, we investigate whether managers overreact to climate disasters. In doing so, we provide a more comprehensive view of how disaster-induced shifts in climate risk assessments impact corporate green innovation.

5.1. The Effects of Increased Climate Risk Assessments on Disaster Disclosure

Our analysis relies on the premise that firms adjacent to disaster zones experience heightened climate-risk assessments following these events. While this increase is evident for directly hit firms, it is useful to establish this effect of adjacent firms to support our focus on the informational channel of disasters. Corporate disclosures, in particular mandatory 10-K filings, provide a natural venue through which managers communicate their assessment of material risks (Graham et al., 2005; Loughran and McDonald, 2016; Matsumura et al., 2024), including those related to climate and extreme weather events. We therefore examine whether treated firms located in adjacent counties increase their discussion of climate disasters in their 10-Ks following nearby disaster events.

We capture disaster-related disclosures using two measures: *HasDisasterDisclosure*, an indicator equal to one if the firm's 10-K filing in a given year contains any disaster-related keywords, and *DisasterDisclosure*, which is defined as the count of disaster-related keywords in the firm's 10-K filing divided by the total number of words in the report, multiplied by 10^4 for scale. We then re-estimate Equation (1) with these two variables as dependent variables.

The results are reported in Table 6. In Columns (1) and (2), the dependent variable is *HasDisasterDisclosure*. Across both specifications, the coefficient of interest, *Treat* × *Post* is positive and statistically significant at the one percent level. The estimate in Column (2) with firm-level controls suggests that firms are approximately 2.83% more likely to discuss climate disasters in their annual reports. In Columns (3) and (4), *DisasterDisclosure* is the dependent variable. Consistent with our expectations, the coefficient of interest continues to be positive and statistically significant. Thus, the results in Table 6 suggest that treated firms increase their discussion of climate disasters in their corporate disclosures, supporting the notion that these firms exhibit increased climate-risk assessments even without experiencing direct disaster impacts. These results are also consistent with prior findings showing that firms increase their environmental disclosures following nearby natural disasters (Dessaint and Matray, 2017; Huang et al., 2022).

5.2. Spillover Effects of Climate Disasters on Adjacent Counties

A potential threat to our inference is the possibility of economic spillovers. Climate disasters can generate significant local economic disruptions in the directly hit counties and potentially spill over to adjacent counties. If adjacent counties experience an economic recession (or expansion) relative to control counties, our results could be driven by general economic changes rather than by changes in climate risk assessments. Additionally, firms in the adjacent counties may share supply chains or product markets with geographically nearby firms in the disaster zones. If the disaster disrupts the suppliers, customers, or competitors in the directly hit areas, treated firms might innovate in response to these supply-chain or competitive shocks rather than to the climate risk information revealed by the disaster itself.

To mitigate these concerns, we conduct two sets of tests. First, we augment our baseline model (Equation (1)) with a comprehensive set of county-level economic indices to capture local economic conditions. We report the regression results in Table 7. In Column (1), we include controls for the level of new business entry (*BusinessEntry*) and exit (*BusinessExit*) in the firm's county in a given year, the number of job creation (*EmploymentCreation*) and destruction (*EmploymentDestruction*), as well as unemployment labor force (*Unemployment*) to capture the intensity of reallocation and creative destruction within the local economy (Davis and Haltiwanger, 1992; Decker et al., 2014). Additionally, we control for GDP (*GDP*) and average personal income in the firm's county (*PersonalIncome*) to further account for aggregate local economic conditions and demand. All count-like control variables are IHS-transformed. In Column (2), we replace these controls with their rates (*BusinessEntryRate*, *BusinessExitRate*, *EmploymentCreationRate*, *EmploymentDestructionRate*, *GDPRate*, *PersonalIncomeRate,* and *UnemploymentRate*) as an alternative set of controls. In both specifications, the coefficient on the interaction term remains positively significant and quantitatively similar to the effects documented in Table 2. These results are consistent with the notion that potential spillover effects from directly hit counties on adjacent counties' economic conditions do not explain the observed increase in green innovation.[11]

In the second set of tests, we exclude from our analysis treated firms with any economic linkages to firms in disaster zones to ensure that our results are not driven by the propagation of the disasters' direct economic impact through peer networks. We use two definitions of peer

[11] In untabulated analysis, we find the nearby disasters has a weakly negative, rather than positive, impact on adjacent counties' economic conditions. Such adverse effects should hinder, rather than stimulate, green innovation among treated firms, implying that any spillover effects would work against our results.

networks to capture economic linkages: supply-chain peers and product market peers. We identify supply-chain relationships from Compustat and product-market relationships from Hoberg and Phillips (2016). The results are reported in Table 8. In Columns (1) and (2), we exclude treated firms with supply-chain peers and product-market peers directly hit by disasters, respectively. Across both specifications, the coefficient on *Treat* × *Post* remains positive and statistically significant with similar economic magnitudes. These findings indicate that disruptions in peer networks do not explain our green innovation results and, thus, are more consistent with the information channel.

5.3. Do Managers Overreact in Their Assessments of Climate Risk?

We next explore whether managers overreact to climate risk information revealed by disasters. Prior research by Dessaint and Matray (2017) argues that the salience of extreme hurricane strikes causes managers to overreact to liquidity risk, leading them to temporarily increase cash holdings—a behavior that reverses after roughly one year as salience wanes. Although our dependent variable, green innovation, reflects a long-term investment horizon distinct from short-term liquidity management, and our dynamic analysis shows no reversal in green innovation response after one year, it remains possible that managers overestimate the level of climate risk their firms face following a climate disaster. Therefore, in this subsection, we examine whether managers' climate risk assessments reflect a similar transient overreaction or a persistent strategic shift by comparing green innovation responses of treated firms experiencing climate disasters for the first time with those of treated firms that have already experienced at least one prior disaster.

If the disaster effect is driven primarily by salience-induced managerial overreaction, we would expect the response to be concentrated among firms experiencing their first disaster. When similar treatment events repeat, they are no longer perceived to be as salient or novel (Kahneman, 2011), and managers of these firms will have already learned from their first overreaction. Thus, the effects should be muted for firms experiencing subsequent disasters. In contrast, if climate disasters serve as information shocks about long-term climate risk, we would expect firms to continue responding to repeated shocks. Under the information channel, the green innovation response will be similar across both groups of firms.

To test this prediction, we modify Equation (1) by splitting the *Treat* variable into *FirstTimeTreat* and *NonFirstTimeTreat* and interacting them with the *Post* variable. *FirstTimeTreat* is an indicator equal to one for first-time treated firms, and zero otherwise; *NonFirstTimeTreat* is an indicator equal to one for non-first-time treated firms, and zero otherwise. The results are presented in Table 9. We find positive and statistically significant coefficients for both interaction terms. In Column (2), which includes firm-level controls, the effect for first-time treated firms is 0.0231 (t-stat = 2.96) and for non-first-time treated firms is 0.0234 (t-stat = 2.72). Importantly, the difference between these coefficients is statistically insignificant (p-value = 0.9687). Thus, the results do not support the notion that managers merely overreact to the salience of climate disasters; rather, they are consistent with managers rationally updating their beliefs about climate risk.

5.4. Robustness Tests

To validate the robustness of our main findings, we conduct a series of tests that incorporate alternative dependent variables, estimation techniques, and definitions of treated firms. We begin by considering alternative dependent variables, as shown in Panel A of Table 10. In Column (1), we use an indicator variable, *HasGreenPatent*, that takes the value of one if the firm files for any green patent in a given year, and zero otherwise. The coefficient on the interaction term *Treat* ×

Post is positive and significant at the one percent level, indicating that firms are more likely to file for green patents in the post-period. In Column (2), we employ another dependent variable, *PercentGreenPatent*, which is the percentage of new green patents out of all patents the firm files in a given year. Similarly, the interaction term continues to be positive and statistically significant. The estimate suggests that treated firms increase the proportion of green patents in their portfolio by 12.68 percentage points relative to control firms.

In Columns (3) and (4), we consider two relative measures of green patent quality, $GreenCitation^{Relative}$ (the average forward citations of the green patents filed by the firm in a given year, scaled by the average citations of all the patents the firm files in that year) and $EconomicValueGreenPatent^{Relative}$ (the average economic value of the green patents filed by the firm in a given year, scaled by the average economic value of all the patents the firm files in that year). The coefficient of interest continues to be positive and statistically significant, suggesting that firms not only increase their green innovation outputs but also enhance the quality of their green patent. Finally, we apply the Poisson regression as an alternative way to mitigate the skewness of the distribution of our main measure, which captures green patent counts (Cohn et al., 2022; Chen and Roth, 2024). As shown in Column (5), our results are robust to using Poisson regression, corroborating our primary findings.

In our final robustness test, we consider alternative definitions of treated firms. The results are reported in Panel B of Table 10. In Column (1), a firm is treated if its headquarters is in an unaffected county within 50 miles of a directly hit county. In Column (2), a firm is treated if its headquarters is in an unaffected county within 150 miles of a directly hit county. In Column (3), a firm is treated if its headquarters is in an unaffected county that borders a directly hit county. In Column (4), a firm is treated if its headquarters is in one of the ten unaffected counties closest to any county directly hit by a disaster. Consistent with our main analysis, the *Treat* × *Post* coefficient is positive and statistically significant across all specifications.

Overall, our robustness tests across various alternative specifications support our central conclusion: firms experiencing heightened climate-risk assessments increase green innovation activities, both in quantity and quality, following climate disasters.

6. Conclusion

Given the increased attention to climate risks in recent years, understanding how firms respond to climate risk is an important question. In this study, we leverage nearby climate disasters to examine whether green innovation is one avenue that firms use to manage climate risk. We posit that because customers and investors increasingly demand climate-conscious actions, and because the physical risk from extreme events can negatively impact firm operations, firms have an incentive to invest in green innovation. We find evidence consistent with this conjecture: firms increase their green innovation efforts, both in quantity and quality, following occurrences of climate disasters.

Moreover, our cross-sectional analyses reveal that this effect is concentrated among firms located in areas with higher levels of disaster-related information supply and demand. These findings highlight an information mechanism: firms are more likely to increase their green innovation response when they operate in environments with more effective information flow. We also provide evidence suggesting that economic spillovers from disaster zones, disruptions in peer networks, and salience-induced managerial overreaction are not the main drivers of our results. Overall, our findings help extend our understanding of the conditions under which firms are motivated to invest in green innovation and should be of relevance to policymakers concerned with how to promote more meaningful corporate green innovation.

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Figure 1: Example – Identifying Treated and Control Firms

This figure provides an example of identifying treated and control firms using Hurricane Michael in 2018. The counties shaded in red are those directly hit by the disaster, as identified using SHELDUS data, which considers a county impacted if it reports any kind of damage caused by a disaster. The gray-shaded counties are those for which the minimum distance between their geographical center and the center of any directly hit county is less than or equal to 100 miles. These counties are considered nearby but not directly hit by the disaster. A firm is considered treated if its headquarters is located in a gray-shaded county and considered a control if its headquarters is in an unshaded county. Firms with headquarters in the disaster zone (red-shaded counties) are not considered in the analysis.

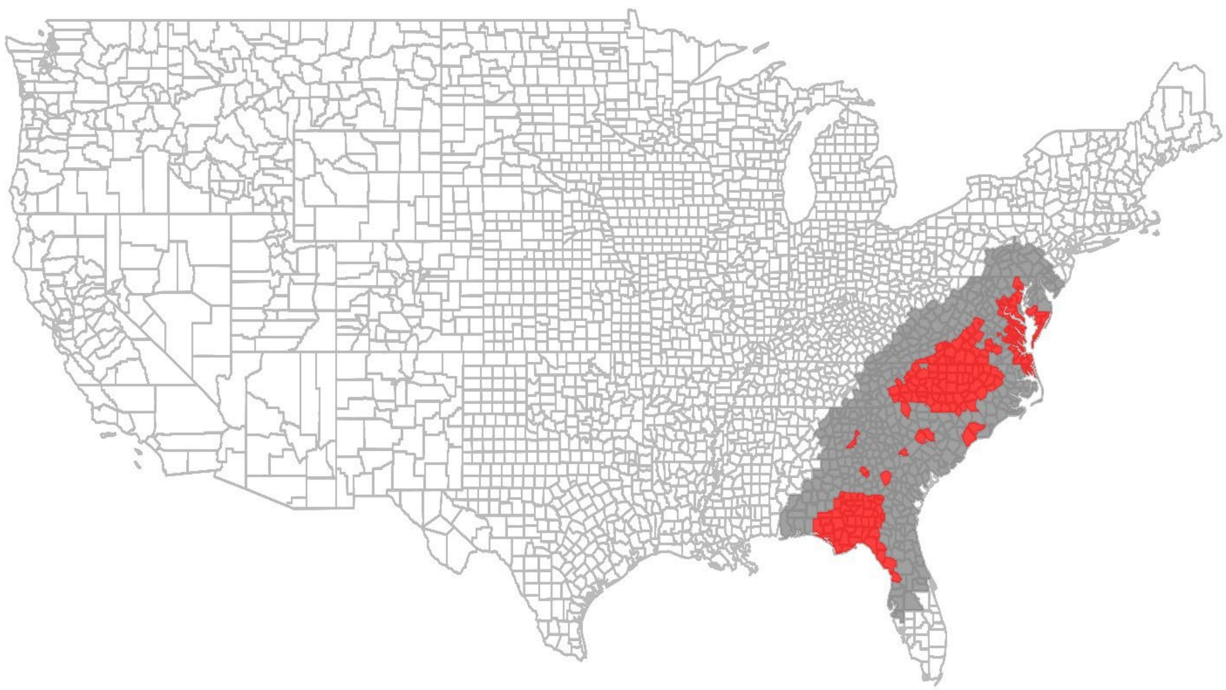


Figure 2: The Dynamic Effects of Increased Climate Risk Assessments on Green Innovation

This figure shows coefficient estimates and 95% confidence intervals from the stacked difference-in-differences regression, estimating the effect of heightened climate risk assessments on green innovation in the period [-5, +5] years around a climate disaster. The disaster occurs at year $t = 0$, with year $t = -1$ serving as the benchmark.

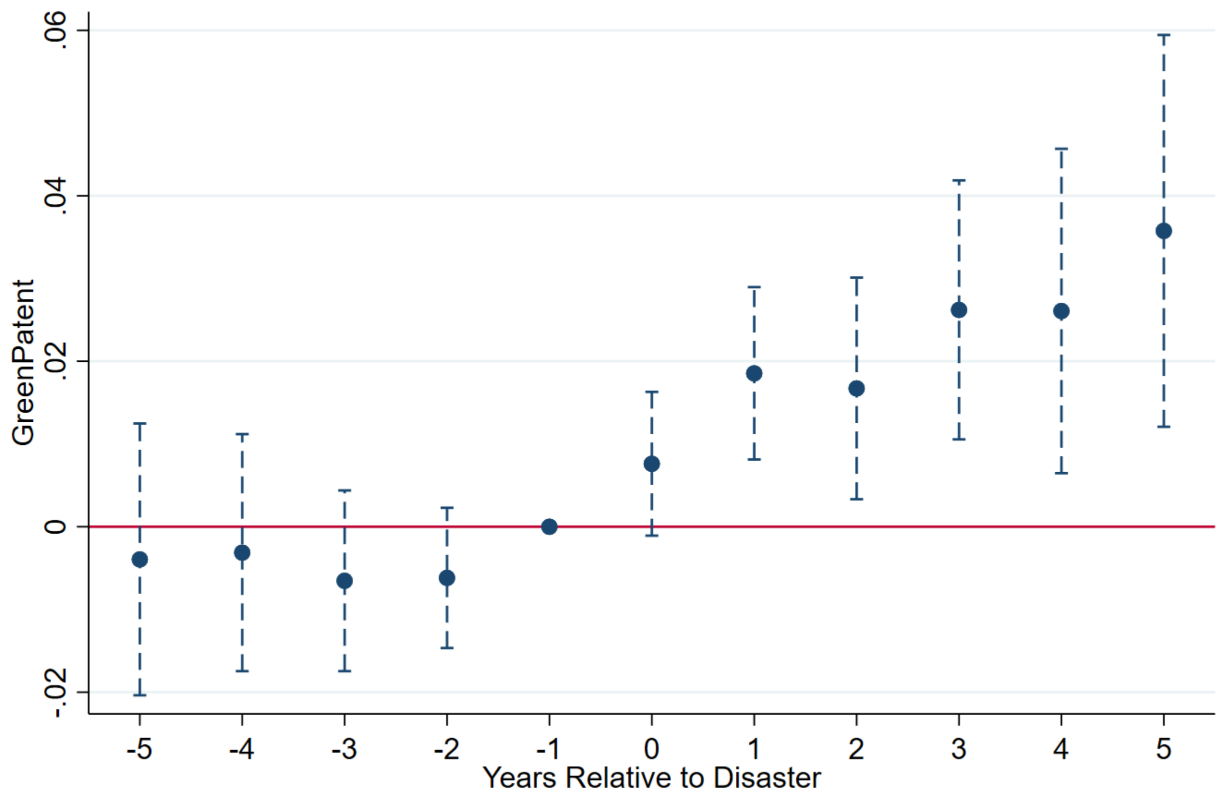


Table 1: Summary Statistics

This table reports summary statistics for variables used in the analysis. All variables are defined in Appendix A. Continuous variables are winsorized at the 1% and 99% levels.

Variable	N	Mean	P25	Median	P75	SD
GreenPatent	461,153	0.155	0.000	0.000	0.000	0.561
Treat	461,153	0.309	0.000	0.000	1.000	0.462
Post	461,153	0.547	0.000	1.000	1.000	0.498
Size	461,153	6.339	4.768	6.150	7.720	2.159
Leverage	461,153	0.173	0.001	0.111	0.276	0.201
ROA	461,153	-0.081	-0.123	0.021	0.077	0.301
Age	461,153	3.265	2.644	3.333	3.872	0.810
Cash	461,153	0.303	0.082	0.236	0.473	0.257
Book-to-Market	461,153	0.504	0.209	0.395	0.687	0.462
FinancialConstraint	461,153	-0.264	-0.341	-0.256	-0.177	0.124
HHI	461,153	0.259	0.127	0.191	0.335	0.198
InstitutionalHolding	461,153	0.500	0.058	0.536	0.883	0.385
CAPX	461,153	0.041	0.014	0.028	0.052	0.042
PreDisasterNews	195,897	15.620	5.426	11.520	20.420	14.230
PreDisasterNewsPerCapita	195,897	0.687	0.244	0.481	1.017	0.828
PreDisasterNewsPerGDP	195,897	0.015	0.006	0.011	0.022	0.013
PreGoogleSearch	195,520	1.355	1.086	1.591	1.724	0.603
PreAnalystQuestions	125,796	0.044	0.000	0.000	0.057	0.090
PreAnalystWords	125,796	0.061	0.000	0.000	0.057	0.164
GreenCitation	461,153	0.185	0.000	0.000	0.000	0.713
EconomicValueGreenPatent	461,153	0.778	0.000	0.000	0.000	3.577
HasDisasterDisclosure	403,256	0.635	0.000	1.000	1.000	0.481
DisasterDisclosure	403,256	0.370	0.000	0.155	0.483	0.593
BusinessEntry	333,923	8.844	8.275	9.001	9.596	0.985
BusinessExit	333,923	8.772	8.186	8.976	9.499	0.978
EmploymentCreation	333,923	12.070	11.490	12.290	12.710	1.015
EmploymentDestruction	333,923	12.000	11.390	12.220	12.690	1.025
GDP	333,923	12.200	11.620	12.400	12.900	1.064
PersonalIncome	333,923	18.740	18.190	18.970	19.430	1.018
Unemployment	333,923	10.510	9.827	10.590	11.200	1.032
BusinessEntryRate	317,695	11.350	10.410	11.270	12.180	1.635
BusinessExitRate	317,695	10.550	9.706	10.360	11.390	1.358
EmploymentCreationRate	317,695	14.670	13.340	14.750	16.060	2.169
EmploymentDestructionRate	317,695	13.880	11.510	12.900	15.540	3.334

GDPRate	317,695	0.035	0.014	0.034	0.060	0.039
PersonalIncomeRate	317,695	0.047	0.025	0.052	0.076	0.041
UnempoymentRate	317,695	5.950	4.223	5.300	7.492	2.291
FirstTimeTreat	461,153	0.040	0.000	0.000	0.000	0.197
NonFirstTimeTreat	461,153	0.269	0.000	0.000	1.000	0.443
HasGreenPatent	461,153	0.092	0.000	0.000	0.000	0.289
FractionGreenPatent (%)	461,153	0.702	0.000	0.000	0.000	3.701
$GreenCitation^{Relative}$	461,153	7.160	0.000	0.000	0.000	29.320
$EconomicValueGreenPatent^{Relative}$	461,153	0.088	0.000	0.000	0.000	0.292

Table 2: Main Results – Increased Climate Risk Assessments and Green Innovation

This table reports estimated coefficients from stacked difference-in-difference regressions estimating the effects of increased climate risk assessments on green innovation. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. The main independent variable of interest is *Treat* × *Post*, where *Treat* is an indicator for treated firms, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent	
	(1)	(2)
Treat* × *Post	**0.0189****	**0.0233*****
	(2.53)	**(2.86)**
Size		0.0306***
		(3.89)
Leverage		0.0196
		(0.75)
ROA		-0.0264**
		(-2.33)
Age		0.0191
		(1.12)
Cash		0.0088
		(0.36)
Book-to-Market		0.0026
		(0.53)
Financial Constraint		0.0502
		(1.50)
HHI		-0.0395
		(-1.10)
Institutional Holding		0.0288
		(1.62)
CAPX		0.0306***
		(3.89)
Cohort-Firm FE	Yes	Yes
Cohort-Year FE	Yes	Yes
Obs.	461,153	461,153
Adj. R^2	0.769	0.769

Table 3: Economic Mechanism: Supply of Disaster-Related Information

This table reports the results of the cross-sectional analyses based on the supply of disaster-related information. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. Column (1) replicates the main result in Table 2 using the reduced sample with news coverage. The main independent variables of interest are the triple interaction terms: *Treat* × *Post* × *PreDisasterNews* (Column (2)), *Treat* × *Post* × *PreDisasterNewsPerCapita* (Column (3)), and *Treat* × *Post* × *PreDisasterNewsPerGDP* (Column (4)). *PreDisasterNews* is the decile rank of a firm's pre-treatment average exposure to disaster-related news, calculated as the average annual count of disaster-related news articles in the firm's location state in the pre-period. *PreDisasterNewsPerCapita* is the decile rank of a firm's pre-treatment average exposure to population-standardized disaster-related news, calculated as the average annual count of disaster-related news articles scaled by the population (in 1,000) of the firm's state in the pre-period. *PreDisasterNewsPerGDP* is the decile rank of a firm's pre-treatment average exposure to GDP-standardized disaster-related news, calculated as the average annual count of disaster-related news articles scaled by GDP (in $1,000,000) of the firm's state in the pre-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent			
	(1)	(2)	(3)	(4)
Treat* × *Post* × *PreDisasterNews	-	**0.0166*** **	-	-
	-	**(3.35)**	-	-
Treat* × *Post* × *PreDisasterNewsPerCapita	-	-	**0.0124*** **	-
	-	-	**(2.70)**	-
Treat* × *Post* × *PreDisasterNewsPerGDP	-	-	-	**0.0113** **
	-	-	-	**(2.51)**
Treat × Post	0.0681***	-0.0047	0.0016	-0.0018
	(3.90)	(-0.27)	(0.08)	(-0.09)
Controls	Yes	Yes	Yes	Yes
Other Interaction Terms	Yes	Yes	Yes	Yes
Cohort-Firm FE	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes
Obs.	195,897	195,897	195,897	195,897
Adj. R^2	0.801	0.801	0.801	0.801

Table 4: Economic Mechanism: Demand for Disaster-Related Information

This table reports the results of the cross-sectional analyses based on the demand for disaster-related information. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. Columns (1) and (3) replicate the main result in Table 2 using the reduced samples with Google search index and conference call, respectively. The main independent variables of interest are the triple interaction terms: *Treat* × *Post* × *PreGoogleSearch* (Column (2)), *Treat* × *Post* × *PreAnalystQuestion* (Column (4)), and *Treat* × *Post* × *PreAnalystWord* (Column (5)). *PreGoogleSearch* is the decile rank of a firm's pre-treatment average level of disaster-related google searches, calculated as the average Google search index of disaster-related keywords in the firm's location state in the pre-period. *PreAnalystQuestion* is the decile rank of a firm's pre-treatment average exposure to analysts' disaster-related questions, calculated as the average number of analysts asking disaster-related questions during conference calls in the pre-period. *PreAnalystWord* is the decile rank of a firm's pre-treatment average exposure to analysts' disaster-related questions, calculated instead as the average total number of disaster-related keywords from analysts' questions during conference calls in the pre-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent				
	(1)	(2)	(3)	(4)	(5)
	Google Search Sample		Conference Call Sample		
Treat* × *Post* × *PreGoogleSearch	**-**	**0.0111****	**-**	**-**	**-**
	-	**(2.10)**	**-**	**-**	**-**
Treat* × *Post* × *PreAnalystQuestion	**-**	**-**	**-**	**0.0103****	**-**
	-	**-**	**-**	**(2.28)**	**-**
Treat* × *Post* × *PreAnalystWord	**-**	**-**	**-**	**-**	**0.0108****
	-	**-**	**-**	**-**	**(2.34)**
Treat × *Post*	0.0680***	0.0232	0.1094***	0.0771***	0.0762***
	(3.88)	(0.91)	(4.31)	(2.98)	(2.95)
Controls	Yes	Yes	Yes	Yes	Yes
Other Interaction Terms	Yes	Yes	Yes	Yes	Yes
Cohort-Firm FE	Yes	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes	Yes
Obs.	195,520	195,520	125,796	125,796	125,796
Adj. R^2	0.801	0.801	0.812	0.812	0.813

Table 5: Increased Climate Risk Assessments and Green Innovation Quality

This table reports estimated coefficients from stacked difference-in-difference regressions estimating the effects of increased climate risk assessments on green innovation quality. The dependent variables are *GreenCitation*, average forward citations of the green patents filed by the firm in a given year with the inverse hyperbolic sine (IHS) transformation applied, and *EconomicValueGreenPatent*, the average economic value of the green patents filed by the firm in a given year. The main independent variable of interest is *Treat* × *Post*, where *Treat* is an indicator for treated firms, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable:			
	GreenCitation		*EconomicValueGreenPatent*	
	(1)	(2)	(3)	(4)
Treat* × *Post	**0.0301****	**0.0440*****	**0.1053****	**0.1192****
	(2.33)	**(3.13)**	**(2.03)**	**(2.31)**
Size		0.0444***		0.0346
		(4.26)		(0.60)
Leverage		-0.0048		0.1074
		(-0.14)		(0.78)
ROA		-0.0568**		-0.0052
		(-2.40)		(-0.08)
Age		0.0835***		0.0734
		(3.43)		(0.70)
Cash		-0.0285		-0.0143
		(-0.83)		(-0.12)
Book-to-Market		-0.0044		-0.0306
		(-0.46)		(-1.08)
Financial Constraint		0.0636		0.3122**
		(0.75)		(1.98)
HHI		-0.0208		0.1883
		(-0.37)		(0.87)
Institutional Holding		0.0738***		0.1650
		(3.14)		(1.40)
CAPX		0.2037		0.4173
		(1.61)		(0.78)
Cohort-Firm FE	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes
Obs.	461,153	461,153	461,153	461,153
Adj. R^2	0.494	0.496	0.611	0.611

Table 6. Increased Climate Risk Assessments and Disaster Disclosure

This table reports estimated coefficients from stacked difference-in-difference regressions estimating the effects of increased climate risk assessments on firms' annual reports over the period [-5, +5] years around a climate disaster. The dependent variables are *HasDisasterDisclosure*, an indicator variable equal to 1 if the firm's 10-K filing mentions any disaster-related keyword and 0 otherwise, and *DisasterDisclosure,* the count of disaster-related keywords mentioned in the firm's 10-K filing, divided by the total number of words in the report, multiplied by 10^4. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX.* Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable:			
	HasDisasterDisclosure		*DisasterDisclosure*	
	(1)	(2)	(3)	(4)
Treat × Post	**0.0403*** **	**0.0283*** **	**0.0546*** **	**0.0364*** **
	(3.93)	**(2.78)**	**(4.30)**	**(2.88)**
Size		0.0348***		-0.0000
		(3.64)		(-0.00)
Leverage		0.0107		-0.0707**
		(0.41)		(-2.35)
ROA		-0.0330*		0.0275
		(-1.86)		(1.35)
Age		-0.0905***		-0.1262***
		(-4.99)		(-4.70)
Cash		0.0075		0.0321
		(0.25)		(1.00)
Book-to-Market		0.0027		-0.0113
		(0.30)		(-0.89)
Financial Constraint		-0.0155		-0.0049
		(-0.22)		(-0.05)
HHI		0.0190		0.0095
		(0.35)		(0.15)
Institutional Holding		-0.0391		0.0160
		(-1.64)		(0.49)
CAPX		0.1073		0.2025
		(1.02)		(1.54)
Cohort-Firm FE	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes
Obs.	403,256	403,256	403,256	403,256
Adj. R^2	0.566	0.568	0.540	0.542

Table 7: Main Results Controlling for County-Level Economic Indices

This table reports the main results controlling for county-level economic indices. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. The main independent variable of interest is *Treat* × *Post*, where *Treat* is an indicator for treated firms, and *Post* is an indicator for observations in the post-period. In Column (1), the local economic indices include *BusinessEntry, BusinessExit, EmploymentCreation, EmploymentDestruction, GDP, PersonalIncome,* and *Unemployment*. In Column (2), the indices instead include *BusinessEntryRate, BusinessExitRate, EmploymentCreationRate, EmploymentDestructionRate, GDPRate, PersonalIncomeRate,* and *UnemploymentRate*. In both columns, *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX* are also included as controls. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent	
	(1)	(2)
Treat × Post	**0.0189****	**0.0380*****
	(2.46)	**(3.19)**
BusinessEntry	0.1386**	
	(2.22)	
BusinessExit	-0.1146*	
	(-1.80)	
EmploymentCreation	0.0619*	
	(1.90)	
EmploymentDestruction	0.0165	
	(0.60)	
GDP	-0.0322	
	(-0.73)	
PersonalIncome	-0.1445**	
	(-2.33)	
Unempolyment	0.0502**	
	(2.05)	
BusinessEntryRate		0.0007
		(0.12)
BusinessExitRate		-0.0204**
		(-2.56)
EmploymentCreationRate		0.0031
		(1.24)
EmploymentDestructionRate		0.0010
		(0.42)
GDPRate		0.0334
		(0.29)
PersonalIncomeRate		-0.0199
		(-0.15)
UnemploymentRate		0.0110**
		(2.29)
Other Controls	Yes	Yes
Cohort-Firm FE	Yes	Yes
Cohort-Year FE	Yes	Yes
Obs.	333,923	317,695
Adj. R^2	0.782	0.785

Table 8: Main Results and the Economic Links to Firms in Disaster Zones

This table reports the estimates of the main analysis excluding treated firms with at least one supply-chain peer (Column (1)) or product-market peer (Column (2)) directly hit by climate disasters. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. The main independent variable of interest is *Treat* × *Post*, where *Treat* is an indicator for treated firms, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent	
	(1)	(2)
Exclusion Criteria:	Supply-Chain Peers	Product-Market Peers
Treat* × *Post	**0.0233*****	**0.0225*****
	(2.99)	**(2.68)**
Size	0.0298***	0.0327***
	(3.70)	(3.44)
Leverage	0.0274	0.0328
	(1.00)	(1.03)
ROA	-0.0249**	-0.0280**
	(-2.15)	(-2.05)
Age	0.0203	0.0252
	(1.17)	(1.24)
Cash	0.0136	0.0138
	(0.54)	(0.46)
Book-to-Market	0.0036	0.0035
	(0.75)	(0.62)
Financial Constraint	0.0525	0.0661
	(1.55)	(1.62)
HHI	-0.0475	-0.0383
	(-1.30)	(-0.91)
Institutional Holding	0.0271	0.0287
	(1.52)	(1.39)
CAPX	0.1378*	0.1792**
	(1.77)	(1.99)
Cohort-Firm FE	Yes	Yes
Cohort-Year FE	Yes	Yes
Obs.	441,001	373,751
Adj. R^2	0.762	0.765

Table 9: First Timers vs. Non-First Timers

This table reports estimated coefficients from stacked difference-in-difference regressions comparing the effects of increased climate risk assessments on green innovation for first-time treated firms and non-first-time treated firms. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. The main independent variables of interest are *Post* × *FirstTimeTreat* and *Post* × *NonFirstTimeTreat*, where *FirstTimeTreat* is an indicator for first-time treated firms, *NonFirstTimeTreat* is an indicator for non-first-time treated firms, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, FinancialConstraint, HHI, InstitutionalHolding,* and *CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent	
	(1)	(2)
Post* × *FirstTimeTreat [a]	**0.0237*** **	**0.0231*** **
	(2.99)	**(2.96)**
Post* × *NonFirstTimeTreat [b]	**0.0184** **	**0.0234*** **
	(2.36)	**(2.72)**
Size		0.0304***
		(3.87)
Leverage		0.0220
		(0.83)
ROA		-0.0252**
		(-2.20)
Age		0.0210
		(1.24)
Cash		0.0130
		(0.52)
Book-to-Market		0.0036
		(0.75)
Financial Constraint		0.0510
		(1.52)
HHI		-0.0393
		(-1.09)
Institutional Holding		0.0277
		(1.56)
CAPX		0.1282*
		(1.66)
p-value ([a] = [b])	0.4802	0.9687
Cohort-Firm FE	Yes	Yes
Cohort-Year FE	Yes	Yes
Obs.	461,153	461,153
Adj. R^2	0.769	0.769

Table 10: Robustness

This table presents the results of the robustness tests for the main analysis. Panel A reports the results using alternative dependent variables and regression specifications. In Column (1), the dependent variable is *HasGreenPatent*, an indicator variable equal to 1 if the firm files a green patent in a year and 0 otherwise. In Column (2), the dependent variable is *PercentGreenPatent*, the number of green patents divided by number of patents filed in a year. In Column (3), the dependent variable is $GreenCitation^{Relative}$, the average forward citations of the green patents, scaled by the average citation of all patents filed by the firm in a year. In Column (4), the dependent variable is $EconomicValueGreenPatent^{Relative}$, the average economic value of the green patents, scaled by the average economic value of all patents filed by the firm in a year. In Column (5), the dependent variable is *#GreenPatent*, the number of green patents a firm files in a year and the Poisson regression is used. Panel B reports the results using alternative definitions of treated firms. In Column (1), a firm is treated if its headquarters is in an unaffected county within 50 miles of a directly hit county. In Column (2), a firm is treated if its headquarters is in an unaffected county within 150 miles of a directly hit county. In Column (3), a firm is treated if its headquarters is in an unaffected county that borders a directly hit county. In Column (4), a firm is treated if its headquarters is in one of the ten unaffected counties closest to any county directly hit by a disaster. Across all specifications, the dependent variable is *GreenPatent*. In both Panels, the independent variable of interest is *Treat* × *Post*, where *Treat* is an indicator for treated firms, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A. Alternative Dependent Variables or Regression Specifications

	Dependent Variable:				
	HasGreenPatent	*PercentGreenPatent*	$GreenCitation^{Relative}$	$EconomicValueGreenPatent^{Relative}$	*#GreenPatent*
	OLS	OLS	OLS	OLS	Poisson
	(1)	(2)	(3)	(4)	(5)
Treat* × *Post	**0.0149***** **(3.64)**	**0.1268**** **(2.40)**	**1.9023***** **(3.81)**	**0.0147***** **(3.57)**	**0.2051**** **(2.15)**
Controls	Yes	Yes	Yes	Yes	Yes
Cohort-Firm FE	Yes	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes	Yes
Obs.	461,153	461,153	461,153	461,153	106,482
Adj. R^2 / Pseudo R^2	0.538	0.396	0.388	0.496	0.800

Panel B. Alternative Definition of Treated Firms

	Dependent Variable: GreenPatent			
	50 Miles	*150 Miles*	*Border Counties*	*Top10Nearby*
	(1)	(2)	(3)	(4)
Treat* × *Post	**0.0212***** **(2.81)**	**0.0231**** **(2.52)**	**0.0176**** **(2.25)**	**0.0213***** **(2.78)**
Controls	Yes	Yes	Yes	Yes
Cohort-Firm FE	Yes	Yes	Yes	Yes
Cohort-Year FE	Yes	Yes	Yes	Yes
Obs.	479,410	458,114	464,835	452,005
Adj. R^2	0.758	0.776	0.764	0.763

Appendix A. Variables Definitions

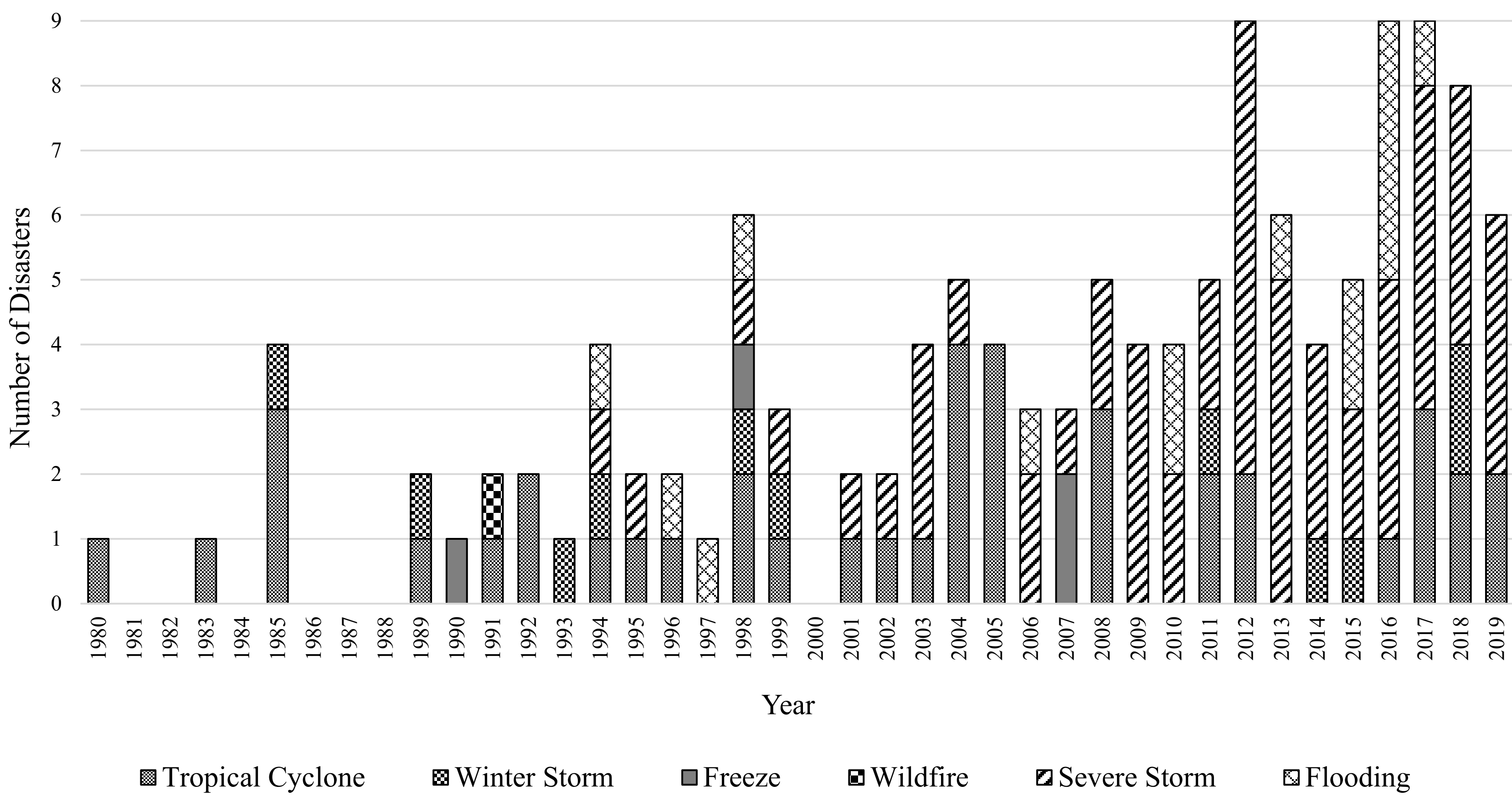
Variables	Definition
Age	The age of the firm (in years) as of a given year with the inverse hyperbolic sine (IHS) transformation applied.
Book-to-Market	The book to market ratio of the firm in a given year, defined as the book equity at the end of a fiscal year (CEQ+TXDB), divided by the market value of equity.
BusinessEntry	The business establishment entries in a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the establishment entry count.
BusinessEntryRate	The business establishment entry rate, defined as the count of establishment entrants in year t divided by the average count of employment active establishments in year t and year t-1.
BusinessExit	The business establishment exits in a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the establishment exit count.
BusinessExitRate	The business establishment exit rate, defined as the count of establishment exits in year t divided by the average count of employment active establishments in year t and year t-1.
CAPX	The amount of capital expenditure of a firm in a given year, divided by total assets.
Cash	Cash and short-term investments of a firm in a given year, divided by total assets.
DirectHit	An indicator equal to 1 for firms located in directly hit counties, and 0 for firms located in counties that are neither adjacent to nor directly hit by the climate disaster
DisasterDisclosure	A text-based measure of the disaster-related disclosure, extracted from the firm's 10-K filing in a given year. It is defined as the count of disaster-related keywords mentioned in the firm's 10-K filing, divided by the total number of words in the report, multiplied by 10^4.
EconomicValueGreenPatent	The average economic value of the green patents filed by the firm in a given year. The economic value of a patent is obtained from Kogan et al. (2017), which is calculated based on capital market reactions to the patent grant.
EconomicValueGreenPatent^{Relative}	The average economic value of the green patents filed by the firm in a given year, scaled by the average economic value of all the patents the firm files in that year. The economic value of a patent is obtained from Kogan et al. (2017), which is calculated based on capital market reactions to the patent grant.
EmploymentCreation	The job creation in a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the absolute number of jobs created.
EmploymentCreationRate	The job creation rate, defined as the count of absolute number of jobs created in year t divided by the average count of employment in year t and year t-1.
EmploymentDestruction	The job destruction in a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the absolute number of job losses.
EmploymentDestructionRate	The job destruction rate, defined as the count of absolute number of jobs destructed in year t divided by the average count of employment in year t and year t-1.

FinancialConstraint	The Whited-Wu index of a firm in a given year.
FirstTimeTreat	An indicator variable equal to 1 if the firm is treated for the first time, and 0 otherwise.
FractionGreenPatent	The number of green patents divided by the total number of patents filed in a given year, multiplied by 100.
GDP	The inflation-adjusted GDP of a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the inflation-adjusted real GDP.
GDPRate	The growth rate of the inflation-adjusted GDP of a county in a given year.
GreenCitation	The average forward citations of the green patents filed by the firm in a given year with the inverse hyperbolic sine (IHS) transformation applied.
GreenCitation^{Relative}	The average forward citations of the green patents filed by the firm in a given year, scaled by the average citations of all the patents the firm files in that year.
GreenPatent	The inverse hyperbolic sine (IHS)-transformed number of green patents that the firm files in a given year. A patent is considered a green patent if at least half of its patent codes are classified as green.
HasDisasterDisclosure	An indicator variable equal to 1 if the firm's 10-K filing mentions any disaster-related keyword and 0 otherwise.
HasGreenPatent	An indicator variable equal to 1 if the firm files a green patent in a given year, and 0 otherwise. A patent is considered a green patent if at least half of its patent codes are classified as green.
HHI	The Herfindahl-Hirschman Index calculated within a firm's four-digit SIC industry in a given year, defined as the sum of an industry's squared market shares (in percentage).
InstitutionalHolding	The average institutional holding ownership of a firm in a given year, calculated using Form 13F holding data.
Leverage	The leverage of the firm in a given year, defined as total liabilities divided by total assets.
NonFirstTimeTreat	An indicator variable equal to 1 if the firm is treated not for the first time, and 0 otherwise.
PersonalIncome	The total personal income of a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the total personal income.
PersonalIncomeRate	The growth rate of the total personal income of a county in a given year.
Post	Indicator variable equal to 1 for the years after the disaster occurrence, and 0 for the years before the disaster, within a [-5,+5] year window around the event.
PreAnalystQuestions	The decile rank of a firm's pre-treatment average exposure to analysts' disaster-related questions, calculated as the average number of analysts asking disaster-related questions during conference calls in the pre-period.
PreAnalystWords	The decile rank of a firm's pre-treatment average exposure to analysts' disaster-related questions, calculated as the average total number of disaster-related words from analysts' questions during conference calls in the pre-period.
PreDisasterNews	The decile rank of a firm's pre-treatment average exposure to disaster-related news, calculated as the average annual count of disaster-related news articles in the firm's state in the pre-period.

PreDisasterNewsPerCapita	The decile rank of a firm's pre-treatment average exposure to population-standardized disaster-related news, calculated as the average annual count of disaster-related news articles scaled by the population (in 1,000) in the firm's state in the pre-period.
PreDisasterNewsPerGDP	The decile rank of a firm's pre-treatment average exposure to GDP-standardized disaster-related news, calculated as the average annual count of disaster-related news articles scaled by GDP (in $1,000,000) in the firm's location state in the pre-period.
PreGoogleSearch	The decile rank of a firm's pre-treatment average level of disaster-related Google searches, calculated as the average Google search index of disaster-related keywords in the firm's location state in the pre-period.
ROA	The return on assets of the firm in a given year, defined as net income divided by total assets.
Size	The inverse hyperbolic sine (IHS) transformation of the firm's total assets in a given year.
Treat	An indicator variable equal to 1 if the firm's headquarters is located in a nearby county that is adjacent to, but not directly hit by, a climate disaster (treated firms), and 0 if the headquarters is located in a county that is neither adjacent to nor directly hit by the climate disaster. A county is considered nearby if its distance to the closest directly hit county is less than or equal to 100 miles. If this distance is more than 100 miles, the county is considered neither adjacent to nor directly hit.
Unemployment	The average unemployment number in a county in a given year, calculated as the inverse hyperbolic sine (IHS) transformation of the absolute average number of unemployment reported.
UnemploymentRate	The unemployment rate, defined as the absolute average number of unemployment reported in year *t* divided by the average count of total labor force.

Appendix B. Distribution of Climate Disasters

This appendix presents the yearly distribution of climate disasters with total monetary damages of at least 1 billion dollars (adjusted to 2019 dollars) used in the analysis.



Appendix C. The Effects of Climate Disasters on Directly Hit Firms

This table reports estimated coefficients from stacked difference-in-difference regressions estimating the effects of direct disaster impact on green innovation. The dependent variable is *GreenPatent*, the inverse hyperbolic sine (IHS)-transformed number of green patents a firm files in a given year. The main independent variable of interest is *DirectHit* × *Post*, where *DirectHit* is an indicator equal to 1 for firms located in directly hit counties, and 0 for firms located in counties that are neither adjacent to nor directly hit by the climate disaster, and *Post* is an indicator for observations in the post-period. Control variables include *Size, Leverage, ROA, Age, Cash, Book-to-Market, Financial Constraint, HHI, Institutional Holding, and CAPX*. Appendix A contains all variable definitions. Cohort-Firm fixed effects and Cohort-Year fixed effects are included to absorb time invariant firm-specific characteristics and time trends in green innovation, respectively. Each cohort corresponds to a disaster. Standard errors are clustered by firm, and *t*-stats are presented in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent Variable: GreenPatent	
	(1)	(2)
DirectHit* × *Post	**-0.0024**	**-0.0003**
	(-0.33)	**(-0.04)**
Size		0.0265***
		(4.57)
Leverage		0.0204
		(1.28)
ROA		-0.0170*
		(-1.95)
Age		0.0085
		(0.75)
Cash		0.0176
		(1.28)
Book-to-Market		-0.0021
		(-0.58)
Financial Constraint		-0.0040
		(-0.10)
HHI		0.0129
		(0.49)
Institutional Holding		0.0300***
		(2.61)
CAPX		0.1161*
		(1.95)
Cohort-Firm FE	Yes	Yes
Cohort-Year FE	Yes	Yes
Obs.	411,157	411,157
Adj. R^2	0.773	0.774

Appendix D. List of Disaster Keywords

acid rain
atmospheric drought
atmospheric river
avalanche
biological disaster
blizzard
bomb cyclone
bombogenesis
bushfire
category 1 storm
category 2 storm
category 3 storm
category 4 storm
category 5 storm
climate disaster
cloudburst
coastal erosion
coastal flooding
coastal inundation
coldwave
convective storm
cyclone
derecho
disast
disaster
disaster aid
disaster area
disaster finance
disaster management
disaster prepar
disaster recovery
disaster relief
disaster risk investment
disaster struck
downburst
drought
dust storm
earthquake
environmental disaster
extreme cold
extreme drought
extreme heat
extreme precipitation
extreme rainfall
extreme weather
firestorm
flash drought
flash flood
flood
flooding
forest fire
freeze
frost drought
geological disaster
glacier melt
hail
hailstorm
heat wave
high wind
hurricane
hydrological disaster
ice melt
ice storm
inland flooding
landslide
lightning
megafire
meteorological disaster
mudslide
natural disaster
natural hazard
nor’easter
polar vortex
precipitation extremes
raging fire
rainstorm
river flooding
rockslide
sandstorm
seasonal drought
severe storm
severe weather
snowstorm
soil erosion
storm
storm surge
stormwater
supercell storm
superstorm
thunderstorm
tidal flooding
tidal wave
tornado
torrential rain
tropical cyclone
tropical storm
tsunami
twister
typhoon
urban flooding
volcanic eruption
volcano
wildfire
wildland fire
winterstorm